

Data Mining

Block 6 - Neural networks & responsible ML

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Today, hour by hour

- | | | |
|---|--|---------|
| ➊ | Recap - one model family left | ~5 min |
| ➋ | Neural networks - from perceptron to MLP | ~40 min |
| ➌ | Training realities - loss curves, leashes, learning rates | ~25 min |
| ➍ | The verdict - MLP vs GBDT vs the baseline | ~15 min |
| ➎ | Responsible ML - fairness, law, and consequences | ~40 min |
| ➏ | Defence & exam briefing - next week, in detail | ~20 min |
| ➐ | Lab - notebook 06_neural_networks | ~35 min |

Breaks roughly every 50 minutes. Last teaching block - next week: defences and the exam.

Last week in five sentences

- 1 Baskets: **support** for materiality, **confidence** for the promise, **lift** for whether anything is there.
- 2 **Apriori's** downward closure is why mining terminates.
- 3 Text becomes columns via **bag-of-words and TF-IDF** - then the whole course applies.
- 4 Vocabulary and idf are fitted parameters - **the golden rule follows text too**.
- 5 Deliverables ship with **honesty clauses**: lift over confidence, a human queue behind every router.

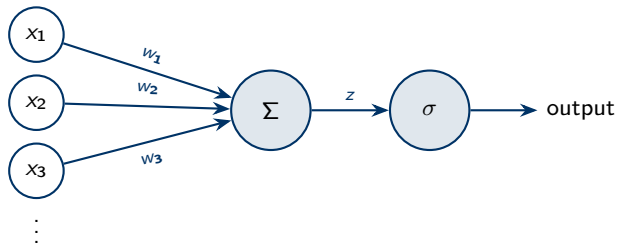
Standings on the real portfolio: logistic **0.731**, GBDT **0.773**. One challenger left - can it take the tabular crown?

The one model family everyone has heard of

- Neural networks power the things your relatives ask about: ChatGPT, image recognition, translation, speech.
- This block builds the **foundations honestly**: what a network is, how it trains, when it helps - and when it is the wrong tool wearing a fashionable name.
- Spoiler discipline: it enters the same tournament as everyone else, against the same baseline, on the same honest numbers.

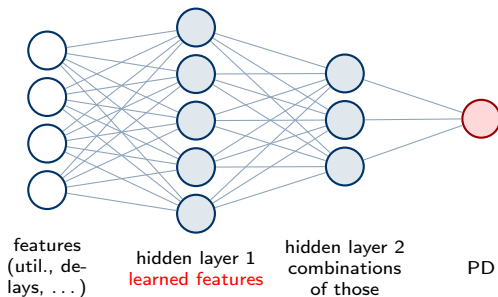
A note on history: the perceptron is from **1958** - older than CRISP-DM, credit scoring, and your instructor combined. Two “AI winters” later, data and GPUs made it bloom. The mathematics barely changed.

The neuron: something you already know



- Weighted sum, squashed: $\sigma(w_0 + w_1x_1 + \dots)$. You met this in Block 2 - one neuron with a sigmoid is logistic regression.
- Everything that follows is a strategy for wiring many of these together.

Stack and repeat: the multilayer perceptron



- Each hidden unit computes its own little logistic-like score of the layer below; the next layer combines *those*.
- The network **engineers its own features** - Block 1's craft, automated. That is the entire selling point.

Why depth at all: the XOR lesson

- A single neuron is a linear cut - and famously **cannot learn XOR** (“either-or”): no line separates the diagonal corners. That 1969 observation helped freeze the field for a decade.
- One hidden layer fixes it: two units each draw a line, the output combines them - the network builds *either-side-of-both* as a **learned feature**.
- Credit translation: “risky iff *high utilisation XOR has savings buffer*” - patterns whose pieces are individually uninformative. Trees find these too (B3); linear models need them hand-fed (B2’s interactions).

Theory footnote: one wide-enough hidden layer can approximate any reasonable function - *can* in principle, not *will* from finite data. The U-curve is not repealed.

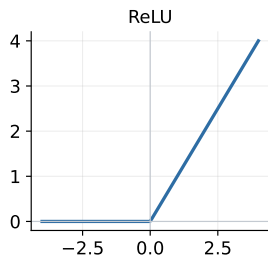
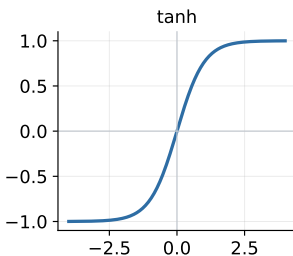
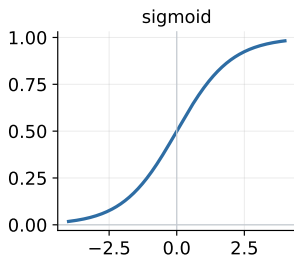
Counting the knobs (worked example)

Our lab network on 10 features: layers $10 \rightarrow 32 \rightarrow 16 \rightarrow 1$.

Layer	weights	biases
input $\rightarrow$ hidden 1	$10 \times 32 = 320$	32
hidden 1 $\rightarrow$ hidden 2	$32 \times 16 = 512$	16
hidden 2 $\rightarrow$ output	$16 \times 1 = 16$	1
total	897 parameters	

- Logistic regression solved the same task with 11. On our 21,000 training rows, 897 knobs are comfortable - the same count on Block 1's little 2,800-row world needed a much tighter leash. Size the leash to the data.
- Modern LLMs: the same counting exercise, with $\sim 10^{11}$ knobs - and correspondingly astronomical data appetites.

Activations: the bend between the layers



- Without a nonlinearity, stacked layers collapse into one big linear model - **the bend is where the power lives**.
- Modern default: **ReLU** ($\max(0, z)$) - cheap, and its gradient doesn't vanish for large inputs the way sigmoid's does. Sigmoid survives at the *output*, where we need a probability (Block 2 says hello).

Training: backpropagation without the tears

- 1 **Forward**: push a batch of customers through; get predicted PDs.
- 2 **Loss**: score them with **log-loss** - the very same loss as logistic regression (Block 2, again).
- 3 **Backward**: the chain rule attributes a share of the error to *every weight* - that attribution is backpropagation, and it is bookkeeping, not magic.
- 4 **Step**: nudge each weight downhill a little (**gradient descent**); repeat for many epochs:

$$w \leftarrow w - \eta \frac{\partial L}{\partial w} \quad (\eta = \text{learning rate})$$

Unlike OLS there is no closed form and unlike logistic regression the landscape isn't convex - many valleys, so **initialisation and seeds matter** (reproducibility rule, weights edition).

Stochastic gradient descent: learning in batches

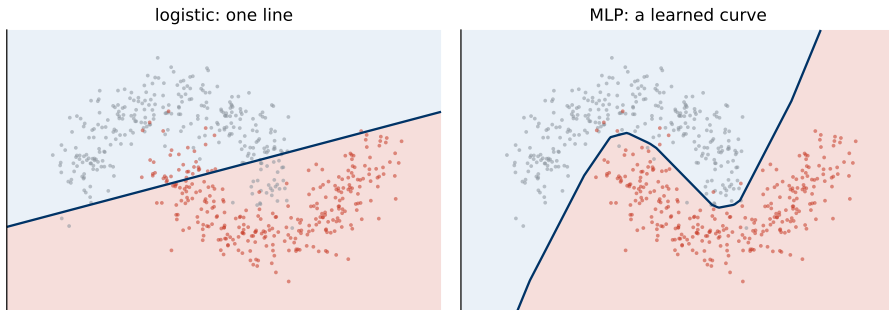
- Computing the exact gradient means touching *every* customer per step - accurate, slow, and pointless at scale.
- **SGD**: estimate the gradient on a random **mini-batch** (say 200 customers), step, repeat. One pass through all batches = one **epoch**.
- Noisy steps are a feature: the jitter helps escape bad valleys of the non-convex landscape - controlled chaos, like bootstrap's controlled resampling (B3).
- **Adam** (the default optimiser) adds per-weight adaptive step sizes and momentum - fewer tantrums, **but it is still gradient descent with a learning rate** you own.

Output layers: one task, one head

Task	Output layer	Loss	Course tie
Binary PD	1 unit, sigmoid	log-loss	Blocks 2–3
Multiclass routing	5 units, softmax	cross-entropy	Block 5
Regression (LGD)	1 unit, linear	squared error	Block 2

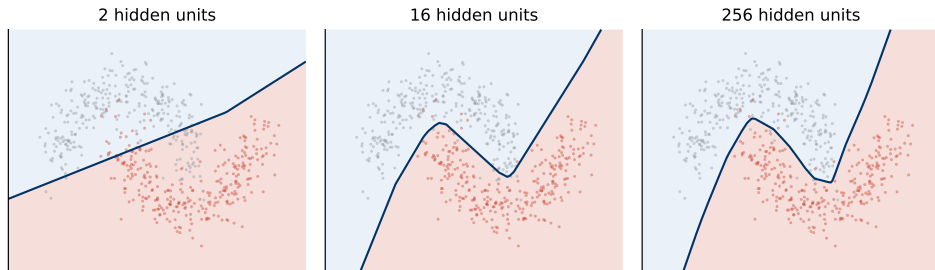
- Softmax = the multi-way sigmoid: five scores in, five probabilities out, summing to one - the complaint router's native output head: $p_k = e^{z_k} / \sum_j e^{z_j}$.
- The body of the network is task-agnostic feature machinery; **only the head and the loss change**. That one design idea carries all of deep learning.
- It also means one body can serve *several* heads at once (PD and LGD from shared learned features) - multi-task learning, mentioned so the term has a home.

What the bend buys: moons, revisited



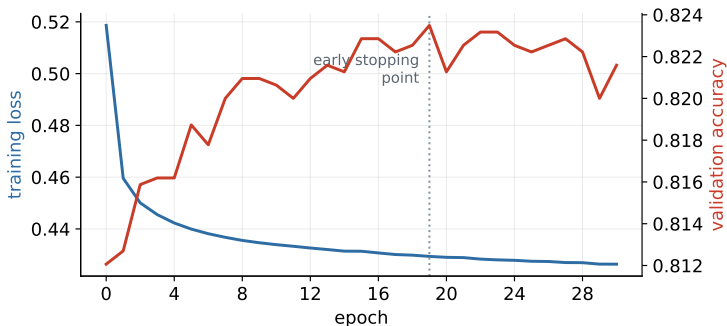
- Block 3's crescents, third visit: logistic must draw a line; the MLP **learns the curve** - no hand-engineered features, no axis-aligned rectangles.
- Smooth learned boundaries are the MLP's signature - compare the tree's staircase.

Capacity: the U-curve, neural edition



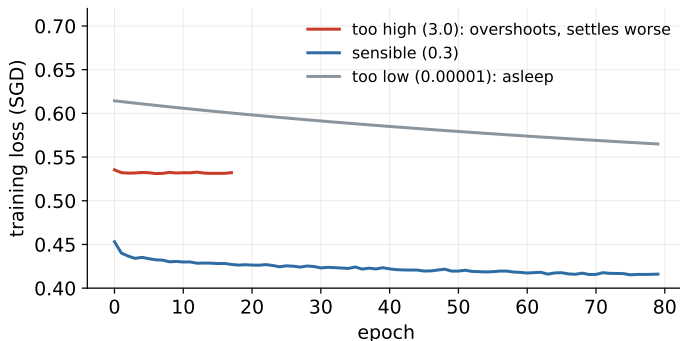
- Two hidden units underfit; 256 memorise the noise along the edge. **Width and depth are the capacity dials.**
- You knew this was coming: same U-curve as polynomial degree, tree depth, and $1/\alpha$. Nothing about neural networks repeals Block 2.

Watching a network learn



- Training loss falls forever; **validation accuracy** rises, peaks, and decays as memorisation sets in - **early stopping** halts training at the peak: the network's everyday leash.
- This one plot is your first diagnostic for any training run - learn to read it before you learn any architecture.

The learning rate: one knob to rule your evening



- Too high: the steps overshoot the narrow valley and settle at a **worse** loss. Too low: the network is asleep. The usable band spans **orders of magnitude** - search it on a log scale (Block 3's tuning advice applies).
- GBDT's learning rate is the same concept - humble steps, more of them.

The leash inventory (where is its leash? - answered)

Leash	Mechanism	Course cousin
Early stopping	stop at the validation plateau	GBDT rounds (B3)
Weight decay (L2)	penalise large weights	ridge (B2)
Dropout	randomly silence units while training	bagging-ish (B3)
Smaller network	fewer units/layers	tree depth (B3)
More data	the only leash that adds signal	- everyone's

- Every mechanism is new; **no idea is new**. Ask the fixed question - where is the leash? - and any future architecture becomes familiar in an afternoon.

Practicalities that bite

- **Scale your features** - gradient descent on unscaled inputs limps; the pipeline with `StandardScaler` is mandatory again (penalties country, not trees country).
- **Seeds**: random initialisation $\Rightarrow$ different runs, different networks; fix `random_state`, expect run-to-run wobble anyway.
- **Missing values**: no native handling - back to Block 1's impute-plus-indicator. (GBDT stays more convenient here.)
- **Cost**: minutes-not-milliseconds training, more tuning surface, hungrier for data - all of it must be **paid for by accuracy** we can measure.

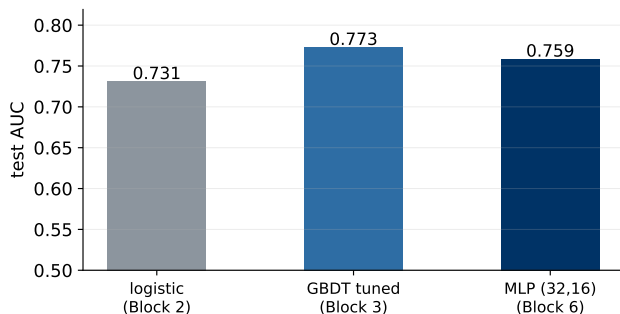
sklearn's `MLPClassifier` is enough for foundations; industry reaches for PyTorch when architectures get custom.

Debugging a training run (field checklist)

- ❶ **Loss is NaN / exploding** → learning rate too high; drop it $10\times$.
- ❷ **Loss flat from epoch one** → rate too low, features unscaled, or a dead ReLU layer.
- ❸ **Train great, validation ugly** → the U-curve says hello: tighten a leash (early stopping, decay, width).
- ❹ **Can it overfit 100 rows?** - the classic smoke test: a healthy network memorises a tiny sample **perfectly**; if it cannot, the bug is in your pipeline, not your patience.
- ❺ **Run twice** - seed wobble bigger than your claimed improvement means you claimed noise (CV spread, B3).

Note the shape of this list: it is Block 1's read-the-traceback discipline, upgraded to a stochastic system.

The final tournament



- MLP: **0.759** - clears the baseline by +2.8 points (complexity **earned**) yet still trails the tuned GBDT (0.773).
- So the real portfolio's final order: $\text{GBDT} > \text{MLP} > \text{SVM} \approx \text{logistic}$. The most fashionable family takes **second place** - the tabular crown stays with boosted trees, exactly what the benchmark literature predicts.

When do neural networks actually win?

- **Perception & sequence data** - images, speech, text, time series: layers exploit **structure** (pixels near pixels, words after words) that tabular rows simply lack.
- **Huge tabular data with interactions** - millions of rows, thousands of features: sometimes; benchmark studies still crown **GBDT** on most mid-size tabular problems.
- **Entity embeddings** - thousands-of-category features (merchant IDs) learned into dense vectors: a genuine tabular win, and the same idea behind Block 5's word embeddings.
- **Representation reuse** - autoencoders (Block 4's reconstruction lens, nonlinear edition), pretrained models, transfer.

Rule of the shop, final form: logistic → GBDT → deep, each step only **after** the previous one is beaten on honest numbers.

What we skipped on purpose (and where it lives)

- **CNNs** - convolutions exploit “pixels near pixels”; vision’s workhorse.
- **RNNs** → **Transformers** - sequence models; attention (2017) is why LLMs exist. Block 2’s “why these models” slide and this table are cousins, not rivals.
- **Autoencoders** - Block 4’s compress-and-reconstruct lens, nonlinear; used for anomaly scores (reconstruction error on a weird customer is high).
- **Pretraining & transfer** - borrow representations learned elsewhere; the economics behind embedding APIs (B5).

Foundations were the deliverable: with the neuron, the loss, SGD and the leash inventory, each of these is one focused week of reading - not a new religion.

The interpretability bill

- Block 2's receipt was **exact and free**. Trees gave a path. The MLP's weights are thousands of numbers with **no individual meaning** - explanation becomes reconstruction (SHAP and friends), approximate and costly.
- In regulated lending the bill is concrete: adverse-action reasons, model validation teams, regulator Q&A - each harder, slower, more contestable.
- So the trade is not "accuracy vs beauty"; it is accuracy vs **auditability, monitoring cost, and legal risk** - priced per use case.

A 0.5-point AUC gain that costs your explanation is a bad trade at a bank and a fine trade in ad-click ranking. **Context prices it.**

Choosing the model family: the course's one table

	logistic	tree	SVM	GBDT/RF	MLP
Interactions	by hand	native	kernel	native	native
Scaling needed	yes	no	yes	no	yes
Missing values	impute+flag	impute	impute	native (Hist)	impute+flag
Explanation	exact receipt	the path	hard	SHAP approx.	SHAP approx.
Probabilities	native	chunky	calibrate	good	native
Tuning burden	tiny	small	moderate	moderate	large
Small data	strong	weak	good	good	weak
Typical tabular rank	baseline	-	mid	top	varies
Regulator comfort	high	high	low	moderate	low

Print this one. Every “which model should we use?” meeting is this table plus costs.

Why this section exists

- A wrong sales forecast wastes money. A wrong **credit model** declines the wrong people, systematically, at scale, silently - the social consequences the course outcomes (K3) name explicitly.
- Every earlier block planted a piece: selection bias (B1), probabilities that price people (B2–3), segments that must not decide (B4), PII in text (B5).
- This section assembles them - on **our own model's numbers**, not hypotheticals.

The operating question is never “is the model accurate?” - it is “**what does the model do to people, and who checked?**”

The legal floor (EU, sketched)

- **GDPR Art. 22:** individuals may not be subject to purely automated decisions with legal effect - credit is the textbook case - without safeguards: **human review on request, meaningful information about the logic.**
- **Duty to state reasons:** declined customers are informed of the grounds - true ones (Block 1's punchline was never just about imputation hygiene; the US calls this "adverse action").
- **Protected attributes:** sex, ethnicity, and friends are off-limits for credit decisions - which is precisely why we audit **by sex** on the next slides; age sits in a grey, jurisdiction-dependent zone (lab exercise).
- **EU AI Act:** credit scoring is listed **high-risk** - documentation, monitoring, human oversight become statutory, not best-practice.

Sketch, not legal advice - the real thing is done **with lawyers in the room**. Your job is to know when to invite them: earlier than feels natural.

Where bias enters the pipeline

Door	Mechanism	Course tie
Labels	“default” reflects <i>past</i> collection practices	B1
Selection	approved-only data; the rejected are absent	B1
Features	proxies encode group membership	today
Measurement	income verified differently across segments	B1
Objective	optimising average cost ignores who pays it	B3
Feedback	the model’s own decisions write next year’s data	next slide

None of these doors is opened by malice. All of them are opened by **default**, unless someone checks.

Feedback loops: the model writes its own future data

- Decline a group more often $\Rightarrow$ fewer loans observed $\Rightarrow$ **no evidence the declines were wrong** $\Rightarrow$ next year's model, trained on the new data, declines them again - with better "justification".
- This is Block 1's selection bias, **motorised**: the model doesn't just inherit the bias, it manufactures more of it.
- Breakers: reject inference (B1's vocabulary), deliberate exploration (approve a small random slice near the boundary - expensive, informative), and monitoring group gaps over *vintages*, not snapshots.
- Any system that acts on its own predictions has one of these loops. Finding it is part of the job now.

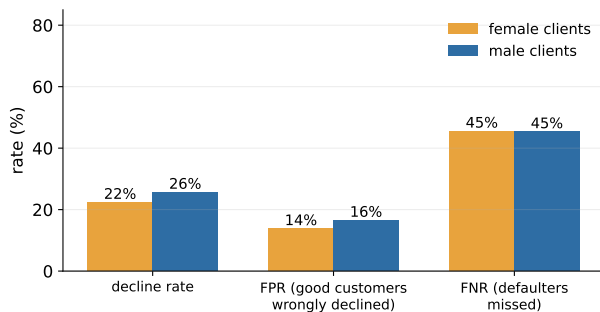
Fairness has competing definitions

For a protected group g and decisions $\hat{y}$:

Criterion	Demands equal. . .	In credit terms
Demographic parity	$P(\hat{y}=1 \mid g)$	decline rates
Equalised odds	FPR and FNR per group	wrongly-declined & missed rates
Calibration by group	$P(y=1 \mid \hat{p}, g)$	a 0.3 means 0.3 for everyone

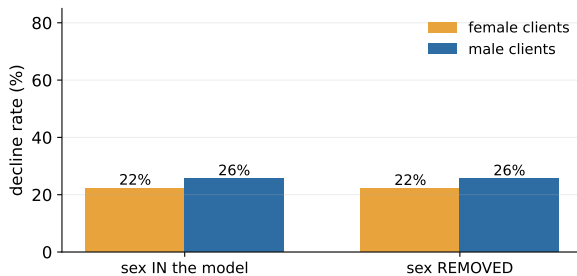
- All three sound obviously right. **Mathematically, they cannot hold at once** when base rates differ between groups (Kleinberg et al., Chouldechova, 2016–17).
- So “make it fair” is not a specification - *choosing which fairness* is the decision, and it is not the data scientist’s alone.

Our own model, audited by sex



- GBDT at Block 3's cost-optimal threshold: men declined at 25.6% vs 22.3% for women, FPR 16.5% vs 13.8% - with base default rates of 23.9% vs 21.0%.
- Read honestly: the gaps are modest and track the base-rate difference - this model does not amplify. (It can go the other way: on Block 1's synthetic portfolio, an age audit turned a 3.6 pp base gap into an 18 pp decline gap.) You only know after you measure - which is the point.

“Just remove sex”: the unawareness experiment



- Removal changes **essentially nothing**: every bar is identical with or without the column. The model never needed sex - six months of payment behaviour already carries whatever group structure exists. That is **proxy discrimination** stated at its purest: deleting the label does not delete the information.
- Corollary: “fairness through unawareness” here is free - and toothless. If the measured gaps needed fixing, the fix would have to act on **decisions** (thresholds, constraints), not on the column list.

The mitigation menu (survey level)

- **Pre-processing:** fix the data - rebalance, collect better labels (recall selection bias: the rejected are missing by construction). Removing proxies alone is the unawareness trap we just watched fail.
- **In-processing:** fairness constraints inside training (equalised-odds penalties; monotonic constraints were a mild cousin).
- **Post-processing:** per-group thresholds - effective, transparent, and **legally delicate** (explicitly using the protected attribute to fix its own harm).
- Always available: **measure and publish** the gaps internally, so the trade-off is decided by people entitled to decide it.

Toolkits exist (fairlearn, AIF360) - the hard part was never the code.

Responsibility beyond fairness

- **Drift & monitoring:** models rot as portfolios move (B3); yesterday's fair model can be today's unfair one - monitor **gaps**, not just Gini.
- **Documentation:** features with time-travel audits (B3), imputation recipes (B1), threshold rationale (B3), fairness measurements (today) - if it isn't written down, it doesn't exist at audit time.
- **Human oversight:** confidence bars and human queues (B5) are not UX niceties; under the AI Act they are architecture.
- **Privacy:** PII in text (B5), data minimisation - collect what the decision needs, not what the crawler found.

Who decides: model governance in a bank

- **First line** - you, the builder: measurements, docs, honesty clauses. Everything this course drilled.
- **Second line** - independent **model validation**: rebuilds your results, attacks your assumptions; a good one is a professional adversarial verifier (and a common first job for this course's graduates).
- **Model risk committee** - owns the accept/reject and the trade-offs no analyst should carry alone (which fairness, whose cost).
- **Audit & regulator** - arrive later, read everything; the written trail *is* the model at that point.

The fairness-metric choice from four slides ago lands **here** - with lawyers and the committee, informed by your measurements.

The responsible-ML checklist (take-home)

- 1 Every feature: *when is it known?* and *may we use it?* - two written answers per column.
- 2 Fairness gaps **measured** (decline, FPR, FNR, calibration by group) before go-live and on schedule after.
- 3 Explanation path exists for every declined customer - and it is **true**.
- 4 A human can be reached, and can overrule.
- 5 Monitoring alarms on drift *and* on gap widening.
- 6 The model's owner is a name, not a department.

Six lines; most model-risk findings in the wild are a missing line from this list.

Next week: how the presentation works

- ~6–8 minutes per team + questions: business framing → data work → two model families vs a baseline → honest evaluation → recommendation.
- Upload deadline: before Block 7 starts - all code (a reproducible notebook) *and* the exact presentation you will show. We may run your notebook top to bottom (Restart & Run All has been the habit since Block 1 - this is why).
- **The AI rule:** assistants were allowed all semester, but you answer for every submitted line. Submissions are read for the fingerprints of **unchecked generation** - conventions nobody can explain, claims the notebook does not compute. The more the project feels written *solely* by AI, the **harder the questions** - expect to explain and modify your code live.
- Bring the **honesty kit**: CV spread, leakage audit, threshold argument, limitations paragraph.

Rubric weights, one more time: framing 10 / data prep 15 / modelling 15 / evaluation 20 / interpretation 15 / reproducibility 5 / **presentation & defence 20.**

Defence failure modes (from the graveyard)

- 1 A suspiciously high AUC presented **proudly** (you know the smell now - find the leak before we do).
- 2 Accuracy on an imbalanced target, unexamined.
- 3 Complex model, no baseline - “earned its keep” cannot be claimed without the keeper.
- 4 Segments/rules with no honesty clause (silhouette, lift vs confidence).
- 5 A notebook that only runs in the order it was written in - which is to say, doesn't.
- 6 Recommendation missing: analysis that ends in metrics instead of **a decision**.

The exam (term 0): what to expect

- Second half of Block 7, **120 minutes**. **~40%** conceptual multiple choice; **~60%** short open questions: pick a method for a described problem, interpret a given output. One question from each block's practice slide appears on the paper - revision pays, literally.
- It tests **selection** and **interpretation** - W3/W4 - not derivations: expect "here is a confusion matrix / rule / coefficient / dendrogram - what do you tell the business?"
- Worked-example slides (Gini split, silhouette, the rule arithmetic, odds ratios) are the calibration for the by-hand questions.
- Best study kit: the six "**Block *N* in five sentences**" slides - they are the syllabus, compressed.

Laboratory

Notebook 06_neural_networks - laptops out.

Lab today

Notebook 06_neural_networks

- train an MLP on the PD task, in a pipeline
- read its loss curve; move the learning rate
- run the final tournament yourselves
- audit the model by sex - then by age (exercise) - and try the removal experiment

Project questions welcome

- last chance to ask in person before the presentation
- stuck on something? ask **today**, not the night before the upload

“Done” = your MLP verdict matches your tournament table, and your fairness audit has numbers in it.

Summary

What to take home - from the block, and the course.

The course in five sentences

- 1 Start from the **decision**; translate it into a task; look at the data before any model touches it.
- 2 Judge everything on **unseen data**, with the right metric for the question and money attached to the threshold.
- 3 Every model family has a **leash**, and complexity must **earn its keep** against a baseline - on real data it did, and the baseline was how we measured by how much.
- 4 Structure without labels is found, argued and **named** - honestly, with a noise check.
- 5 Models act on people: **explanations, fairness gaps and human oversight** are deliverables, not decoration.

The subtitle kept its promise: **how not to fool yourself**.

Project status & reminder: the night before the defence

- 1 **Upload everything before class**: all code + the exact presentation. What you upload is what you present.
- 2 **Restart & Run All** - on a machine that isn't yours if you can.
- 3 Every number on your slides **traceable** to a notebook cell.
- 4 One sentence ready per rubric row - especially *interpretation*: what should the business *do*?
- 5 The two questions you fear most - written down, with your best honest answers ("we didn't have time to X" beats improvised fiction).
- 6 Timing: 6–8 minutes is **~6 slides of content**, not 20. Rehearse once, aloud.
- 7 Sleep. A rested "I don't know, but here's how I'd find out" outscores a tired bluff every time.

Exam practice - multiple choice

Q1. The role of early stopping when training an MLP:

- ☐ a) it speeds up each epoch
- ☐ b) it watches a validation score and halts training before the network overfits - the network's everyday leash
- ☐ c) it prevents exploding gradients
- ☐ d) it reduces the parameter count

Q2. Removing sex from the features left every fairness metric essentially unchanged. Why?

- ☐ a) the audit code was broken
- ☐ b) the model was already fair, so nothing could change
- ☐ c) other features carry (proxy) the same group information - deleting the label does not delete the information
- ☐ d) sex was never predictive to begin with

Exam practice - open question

In the style of the term-0 exam (~60% open questions)

On the tabular PD task: logistic 0.731, MLP 0.759, GBDT 0.773 test AUC. Which model do you recommend deploying, and what *besides* AUC belongs in that decision? Name at least three considerations.

4-6 sentences. W3 in its purest form: select and defend.

Next week: defences & exam.

Bring the honesty kit. Good luck - you won't need it.

Keep learning - Block 6 reading

- **LeCun, Bengio & Hinton** - *Deep Learning* (Nature, 2015) - the canonical review article; where today's foundations lead.
- **3Blue1Brown** - the neural-network video series: the best visual account of backpropagation available.
- **Barocas, Hardt & Narayanan** - *Fairness and Machine Learning* (free at fairmlbook.org) - today's fairness section, in full.
- **Rudin** - *Stop Explaining Black Box Models for High Stakes Decisions...* (Nature Machine Intelligence, 2019) - the interpretability bill, argued sharply.